CASE STUDY: RETRO C++ CONTAINER TRACKER

A Minimalist C++ Console Application

A Single-File C++ Project for Teaching Core Concepts

REPOSITORY

github.com/biswyaa28/Shipping-Container-Tracking-System

Problem & Objectives

/// REAL-WORLD CONTEXT

Global logistics require efficient **container tracking systems** to monitor shipments across ports and distribution centers.

/// PROJECT GOALS

- Simplicity-first architecture
- Retro 1980s terminal UI
- Full requirement fulfillment
- Educational value for C++ beginners

RETRO_STORAGE_MEDIA

Methodology

KEY_DESIGN_CHOICES

Single .cpp File

177 lines total - easy to read and understand

Parallel Arrays

No classes/structs - procedural programming approach

C++11 Standard

No external dependencies - portable across platforms

Retro Console UI

Green text, ASCII borders - 1980s terminal aesthetic

```
/// DATA STRUCTURE
string ids[MAX_CONTAINERS];
string types[MAX_CONTAINERS];
string sizes[MAX_CONTAINERS];
string locations[MAX_CONTAINERS];
string statuses[MAX_CONTAINERS];
string history[MAX_CONTAINERS][MAX_HISTORY];
int history_count[MAX_CONTAINERS];
```

CAPACITY

100 Containers

10 History Records

Core Features Demo

SYSTEM_MENU

CONTAINER TRACKING SYSTEM

1. Register Container
2. Update Location
3. Update Status
4. View All Containers
5. View History
6. Generate Report

ENTER CHOICE:

/// WORKFLOW 1: CONTAINER REGISTRATION

ID: CONT001
TYPE: DRY
SIZE: 20FT
LOCATION: NEW YORK
STATUS: IN TRANSIT

✓ CONTAINER REGISTERED SUCCESSFULLY

/// WORKFLOW 2: VIEW HISTORY

1. NEW YORK → Initial Location
2. SINGAPORE → Location Updated

Requirement Mapping

/// COMPLIANCE STATUS

✓ Container Registration	COMPLETE
✓ Location Updates	COMPLETE
✓ Status Updates	COMPLETE
✓ View All Containers	COMPLETE
✓ View History	COMPLETE
✓ Generate Reports	COMPLETE
✓ Retro Console Interface	COMPLETE

TOTAL REQUIREMENTS

7/7

100% COMPLIANCE

CODE METRICS

Lines of Code	177
Files	1 .cpp
Memory	~10 KB

Why It Works for Beginners

PEDAGOGICAL_VALUE

/// KEY ADVANTAGES

- **Easy to Read & Modify**
  Single file, clear structure
- **No OOP Complexity**
  Procedural approach for beginners
- **Clear Function Architecture**
  Modular design with separate functions
- **One-Command Compilation**
  Simple build process

/// COMPILATION COMMAND

```
$ g++ -std=c++11 -o container_tracker
  container_tracker.cpp
```

LINES OF CODE

177

Manageable scope

COMPILE TIME

<1s

Instant feedback

Perfect for: Console I/O, arrays, modular functions

Conclusion & Takeaways

FINAL_SUMMARY

"Simplicity ≠ lack of functionality"

Starter Project

Console I/O, arrays, modular functions

Intentional Constraints

Improve clarity & creativity

Educational Artifact

Foundational C++ concepts

GITHUB REPOSITORY

github.com/biswyaa28/Shipping-Container-Tracking-System

CLONE → COMPILE → LEARN

/// SESSION_END

THANK YOU

Questions & Discussion Welcome

A minimalist approach to learning foundational C++ concepts through practical application.

GITHUB REPOSITORY

github.com/biswyaa28/Shipping-Container-Tracking-System

COMPILATION

```
$ g++ -std=c++11 -o tracker container_tracker.cpp
```

PROJECT STATS

177 Lines • C++11 • Single File

SCAN FOR CODE

[QR CODE PLACEHOLDER]